**Flight Software Workshop Agenda**
**Day 1 - Monday, February 9th 2026, 8:30 am - 4:00 pm**

| Type | Topic/Activity | Presenter | Time | Duration |
|---|---|---|---|---|
| Introduction | 00 Introduction | Jeff Levison | 8:30 | 15 |
| Presentation | 01 Spacecraft Archtecture and System Overview | Tim Canham | 8:45 | 15 |
| Presentation | 02 FSW System Engineering - Capability Layers<br>Services<br>Applications<br>Drivers | Tim Canham | 9:00 | 30 |
| Presentation | 03 FSW Architecture, Requirements & Design<br>Software architecture<br>Software requirements Structural analysis<br>Data flow diagrams<br>Software design<br>State machine models | Garth Watney | 9:30 | 60 |
| Break | | | 10:30 | 15 |
| Presentation | 04 F' Introduction | Tim Canham | 10:45 | 75 |
| Lunch | | | 12:00 | 60 |
| | 05 FPP Introduction | Rob Bocchino | 1:00 | 15 |
| Exercise | Lab project introduction | Michael Starch | 1:15 | 15 |
| | Lab project setup | Michael Starch | 1:30 | 15 |
| | System requirements definition | Michael Starch | 1:45 | 30 |
| Break | | | 2:15 | 15 |
| Presentation | 06 FSW Design<br>FSW system modeling in F' (Types, ports, components, topology definitions)<br>Data types, interfaces<br>Commands, telemetry events, parameters and data products<br>Concurrency, - initialization, memory allocation<br>OS abstraction layer (message queues, mutex, tasks, file system)<br>Interface design patterns<br>Serializing and deserializing data<br>Designing and visualizing topologies | Michael Starch | 2:30 | 90 |

| Day 2 - Tuesday, February 10th, 2026, 8:30 am - 3:30 pm | | | | |
|---|---|---|---|---|
| **Type** | **Topic/Activity** | **Presenter** | **Time** | **Duration** |
| Exercise | Component Design and Initial Implementation<br>Implementing commands and events | T. Boyer-Chammard | 8:30 | 60 |
| Exercise | Initial Component Integration<br>F´ Deployment and introduction to the GDS<br>Establishing the initial topology | T. Boyer-Chammard | 9:30 | 30 |
| Break | | | 10:00 | 15 |
| Presentation | 07 Reducing Risk<br>Defensive coding<br>Avoiding C and C++ pitfalls<br>Static and dynamic analysis | Rob Bocchino | 10:15 | 45 |
| Presentation | 08 Suggestions for Coding Style<br>Modules and components Functions<br>Expressions and statements | Rob Bocchino | 11:00 | 45 |
| Presentation | 09 Data Structures Library Overview<br>Review of basic data structures for FSW<br>Overview of F Prime data structure library | Rob Bocchino | 11:45 | 15 |
| Lunch | | | 12:00 | 60 |
| Exercise | Full Component Implementation<br>Implementing telemetry, parameters, port calls | Kevin Ortega | 1:00 | 60 |
| Exercise | System Integration<br>Completing the topology | Kevin Ortega | 2:00 | 30 |
| Exercise | Running on Hardware | Kevin Ortega | 2:30 | 60 |
| Lab Tour (optional) | | | 3:30 | 60 |

| Day 3 - Wednesday, February 11th, 2026, 8:30 am - 4:00 pm | | | | |
|---|---|---|---|---|
| **Type** | **Topic/Activity** | **Presenter** | **Time** | **Duration** |
| Presentation | 10 Unit Testing<br>Basic principles of testing<br>Writing modular tests<br>Achieving and checking code coverage<br>Picking good inputs | Rob Bocchino | 8:30 | 30 |
| Presentation | 11 F´ Unit Test Framework | Rob Bocchino | 9:00 | 30 |
| Exercise | Writing unit tests | Rob Bocchino | 9:30 | 60 |
| Break | | | 10:30 | 15 |
| Presentation | 12 Systems Testing F' GDS overview<br>Test API | Michael Starch | 10:45 | 45 |
| Lunch | | | 11:30 | 60 |
| Exercise | System testing with the GDS | Celeste Smith | 12:30 | 65 |
| Presentation | 13 FSW Development Process<br>FSW level process - waterfall model, artifacts, reviews<br>Component level process - checklists, reviews<br>FSW planning and reporting | Aadil Rizvi | 1:35 | 60 |
| Presentations | Lightning Talks | Various | 2:35 | 80 |
| Closing | | Jeff Levison | 3:55 | 5 |

| Lightning Talks - Agenda | | | |
|---|---|---|---|
| **Affiliation** | **Title** | **Presenter** | **Time** |
| Johns Hopkins University | OrbitDSP: Integrated Signal Processing and Orbital Dynamics for Embedded Systems | Herbert Beaumont | 2:40 |
| Texas State University | PROVES kit: The Open-Source CubeSat | Saidi Adams<br>Aaron Siemsen | 2:45 |
| JPL | Hermes: A Command & Telemetry Framework | Andrei Tumbar | 2:50 |
| Blue Eye Soft | Flight Software Challenges for Space Weather–Aware Autonomous Systems | Srikanth Kodeboyina | 2:55 |
| Aclectic Systems Inc. | SynthetixTM and PolymathTM, A Work-In-Progress Towards A Digital Twin Authoring Environment and Edge Computing Platform for Spacecraft Systems | Yahya H. Mirza | 3:00 |
| JPL | Auto Pilot | Andrew Johnson | 3:05 |
| Varda Space Industries | Design and Evaluation of a Lightweight On-Orbit Probe for the NASA GPDM Satellite | Atilla Saadat | 3:10 |
| University of Washington | HuskySat-2: Autonomous Optical Navigation for CubeSats | Yaj Jhajhria | 3:15 |
| JPL | HPSC and ENGAGE Avionics Platforms | Eli McMahon | 3:20 |
| Utah State University | Leveraging F' V4 Subtopologies for a Custom Communication Protocol | Boston Larson<br>Gavin Steel<br>Ethan Wayland | 3:25 |
| Yonsei University | CubeSat activities at Yonsei University | Youngho Eun | 3:30 |
| JPL | Rover Operations Center | Jennifer Trosper | 3:35 |
| Cal Poly Pomona | CADENCE: Designing Flight Software for the University Nanosat Program Using F Prime | Kai Vann | 3:40 |
| Oligo Space | Oligo Space - F' in Context | Samuel Villarreal | 3:45 |
| JPL | New JPL Option for Personalized Industry and Academia Support | Benji Donitz | 3:50 |

**Ordering subject to change**

**Alternate Presentations**

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| Flapmax | Bounded Reconfiguration Under Flight Software Authority for Autonomous Small | Dave Ojika |